DIGO BIKAS INSTITUTE

Research Brief
May 2023

CLIMATE INDUCED LOSS AND DAMAGE, ECONOMIC LOSSES:
A CASE STUDY OF HANUMANNAGAR KANKALINI MUNICIPALITY, SAPTARI DISTRICT, NEPAL

Introduction

On 18th - 20th October 2021, in the post monsoon period southern plains of Nepal, Tarai Madhesh received unusual excessive rainfall for three consecutive days resulting to the flood which resulted in heavy loss of the ready to harvest rice that was not collected This was one of the extreme events faced by farmers of Hanumannagar Kankalini Municipality that resulted in the huge loss of crops and paddy.

Four lakhs Thirty-one thousand one hundred sixty-four-hectare land was cultivated in 2078 in Madhesh Province of which forty-eight thousand six hundred eighty-four hectare cultivated land was impacted by rainfall of the October 2022. This accounted 11 % of the total land cultivated with rice.

Table 1: Cultivated rice field in 2021 in different district of Madhesh Province

S.N.	District	Cultivated Rice field	Losses	% Loss
1	Saptari	66150 Ha	16537 Ha	25 %
2	Sarlahi	43000 Ha	7040 Ha	16.37%
3	Siraha	58000 Ha	8700 Ha	15%
4	Bara	58214 Ha	4994 Ha	8.57 %
5	Rautahat	38750 Ha	3155 Ha	8.14%
6	Dhanusha	58000 Ha	4680 Ha	8.06%
7	Mahottari	45500 Ha	3078 Ha	6.76%
8	Parsa	35050 Ha	500 Ha	1.42%

Source: Ministry of Land Management, Agriculture and Cooperative, Madhesh Province

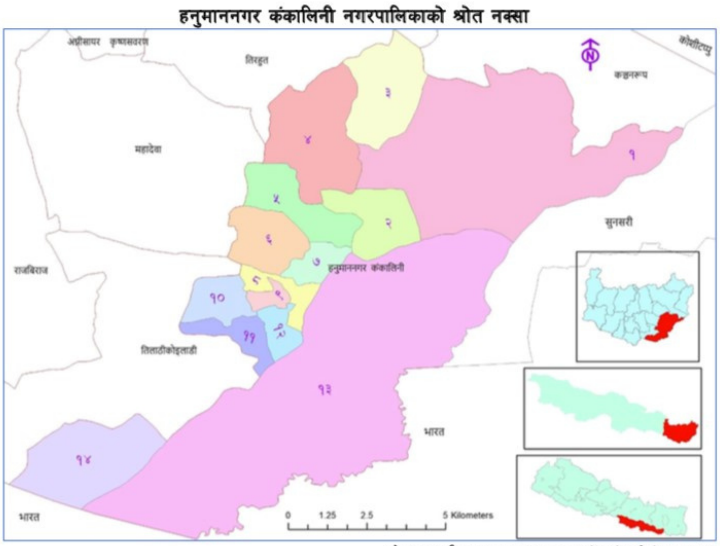
Key Highlights

7250 Tons of Rice has been lost due to the flood which resulted in heavy loss of the ready to harvest rice that was not collected and worth of NPR 18 crore 10 lakh 11 thousand and one hundred only i.e., 1.4 million USD in Hanumannagar Kankalini Municipality, Saptari.

Study Area

Saptari district was most affected district with 25 % loss in the Madhesh Province and Tilathi Koiladi Rural Municipality and Hanumannagar Kankalini Municipality in Saptari[2] was the most affected among the local government as per the Agriculture Knowledge Centre, Saptari. As the Tilathi Koiladi Rural Municipality did not have any segregated wide use detailed data on the loss, the study focused only at the Hanumannagar Kankalini Municipality, Saptari District

हनुमाननगर कंकालिनी नगरपालिकाको श्रोत नक्सा



स्रोत: स्थानीय तह पुनर्संरचना आयोगको प्रतिवेदन २०७४

Table 2: Ward wise population details of the Hanumannagar Kankalini Municipality

WARD	HOUSEHOLD			POPULATION		
	Male	Female	Total	Male	Female	Total
1	1339	145	1484	4408	3396	7804
2	304	38	342	781	678	1459
3	983	82	1065	3172	2502	5674
4	1147	214	1361	3155	3122	6277
5	366	37	403	1649	756	2405
6	678	160	838	1789	1670	3459
7	486	63	549	1179	1094	2273
8	439	48	487	1273	1189	2462
9	293	40	333	984	977	1961
10	349	113	462	1057	987	2044
11	363	43	406	1294	954	2248
12	481	64	545	1441	1199	2640
13	302	36	338	1094	868	1962
14	1513	173	1686	3774	3660	7434

Source: 2075 Household survey

Analysis

2879 farmers including small holder famers were affected by the unusual excessive rainfall in the third week of October, 2021. Ward no 14 was the most affected with 660 famers and ward no. 7 was the least affected with 67 farmers. Total of 1,518 Hector area of land where rice was harvested but not collected was damaged out of this ward no 1 had the most damaged paddy field accounting 614.5 Ha and ward no 7 had the least damage of 31 Hector area of harvested land. Ward no 13, also known as Srilanka Tapu was inaccessible at the time of the data collection so there is no available data for ward no 13.

Table 3: Number of farmers and total harvested land affected in different wards

Ward	No. of Farmers affected	Total land affected in Hector
1	657	614.5 Ha
2	170	109 Ha
3	302	169 Ha
4	143	57 Ha
5	146	50 Ha
6	433	56 Ha
7	67	31 Ha
8	261	65 Ha
9	177	49 Ha
10	269	62.5 Ha
11	173	28 Ha
12	81	25 Ha
13	NA	NA
14	660	201.5 Ha
TOTAL	**2879**	**1518 Ha**

Table 4: ward wise data on the damaged rice in Hanumannagar Kankalini Municipality

Ward	Total tons of rice damaged
1	2945 t
2	522 t
3	807 t
4	272 t
5	237 t
6	269 t
7	147 t
8	311.5 t
9	234 t
10	298.5 t
11	134 t
12	119 t
13	NA
14	963 t
TOTAL	**7250 t**

7250 tons of rice was damaged by 18th-21st October rainfall in Hanumannagar Kankalini Municipality, Saptari District, Madhesh Province. Ward no 1 had the highest damage of 2945 t and ward 11 with 134t had the least damage in the municipality.

Total 7250 Tons of Rice has been lost due to the flood which resulted in heavy loss of the ready to harvest rice that was not collected and worth of NPR 18 crore 10 lakh 11 thousand and one hundred only i.e., 1.4 million USD.

Table 5: ward wise economic losses in Hanumannagar Kankalini Municipality

Ward	Economic losses (NPR)
1	73368000
2	13043000
3	20186200
4	6740600
5	5923600
6	6724000
7	3682300
8	7767200
9	5848800
10	7446400
11	3296000
12	2967000
13	NA
14	24018000
TOTAL	**181011100 (NPR)**

The trends of the rainfall in October time series was calculated using the non-parametric Mann-Kendall (MK) test which has been commonly used to estimate the monotonic trends and their significance in hydro-climatic time series (Poudel et al., 2020). The Mann–Kendall (MK) statistical test compares the relative magnitudes of sample data rather than the data values themselves (Gilbert, 1987), and the advantage of the test is its low sensitivity to abrupt breaks due to inhomogeneous time series (Jaagus, 2006).

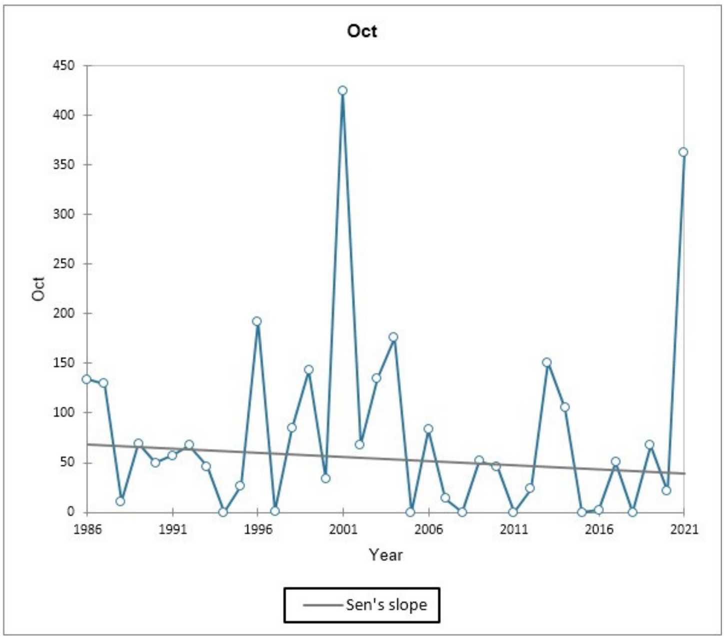


Figure 1: Trend and Magnitude of Rainfall in the Month of October over 30 yrs. in the study area

The results of the Mann-Kendall analysis to deduct the trend and the magnitude of the trend using Sen's slope (Q value) for rainfall trend in the month of October over 30 years in the study area is shown in the Figure. The trend analysis revealed that the annual rainfall in the area over the 30 years for the month of October showed a non-significant decreasing trend with slope of -0.85 at 0.05 significance level. The analysis of the series of rainfall in October indicated that the maximum rainfall was 424.7mm in 2001 followed by 362.72 mm in 2021. The magnitude of rainfall in the years 1994, 1997, 2005, 2008, 2011, 2015 and 2018 is zero for the month of October.

We can see in the graph that the total rainfall in October of year 2001 had a heavy rainfall, which even exceeds the total rainfall of 2021. In 2001, there were nine days of rainfall in October with highest rainfall of two days, 130 and 103mm respectively that accounts for the total maximum rainfall. In addition, there have been a few rainy days below 50 mm in October 2001. However, it is to be noted that it doesn't necessarily mean that in 2001 there have been a very heavy rainfall in a single day like in 2021. In 2021, there was rainfall of 273mm in a single day, which was the highest for any day for the month of October during the 36 years period. This shows that the numbers of rainy days were more (9 days in total) in October of 2001 as compared to October of 2021 where the number of rainy days are few but the intensity is high. In addition, the rainfall of this intensity is unprecedented in the month of October over 36 years of time.

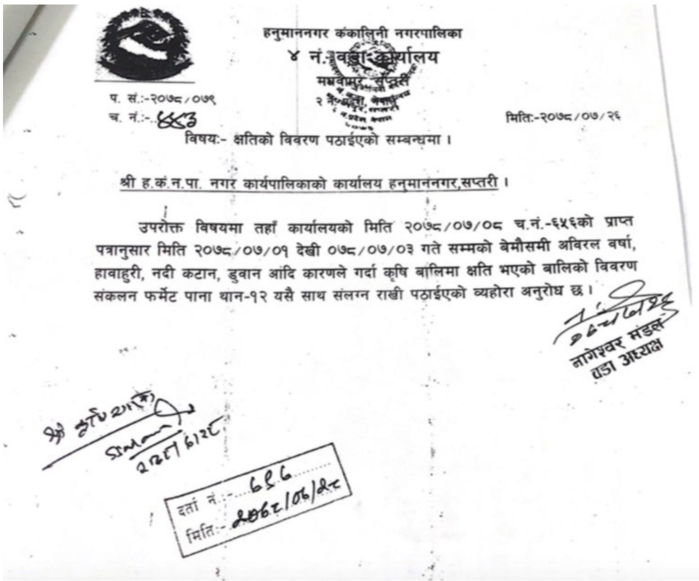
Key Informants Interview During the Research

Regermination of Destroyed Paddy From the Post-monsoon Flood

Data Authentication

Ward wise loss and damage for the Hanumannagar Kankalini Municipality was collected from the Agricultural department at the Hanumannagar Kankalini Municipality.

A specific template was developed based on which ward office and local police unit gathered the information.


हनुमाननगर कंकालिनी नगरपालिका
४ नं. वडा कार्यालय

प. सं.:-२०७८/०७९
च. नं.:- 443

मितिः-२०७८/०७/२६

विषयः- क्षतिको विवरण पठाईएको सम्बन्धमा।

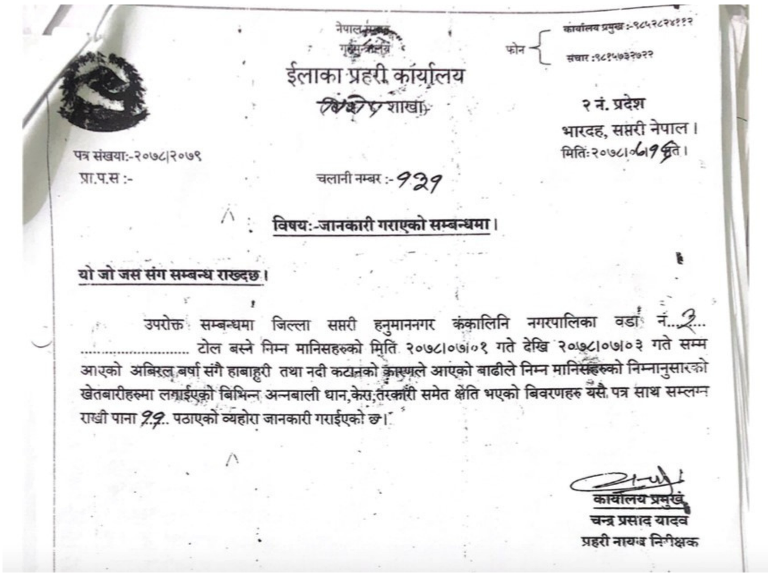
श्री ह.कं.न.पा. नगर कार्यपालिकाको कार्यालय हनुमाननगर,सप्तरी।

उपरोक्त विषयमा तहाँ कार्यालयको मिति २०७८/०७/०८ च.नं.-६५६को प्राप्त पत्रानुसार मिति २०७८/०७/०१ देखी ०७८/०७/०३ गते सम्मको बेमौसमी अविरल वर्षा, हावाहुरी, नदी कटान, डुवान आदि कारणले गर्दा कृषि बालिमा क्षति भएको बालिको विवरण संकलन फर्मेट पाना थान-१२ यसै साथ संलग्न राखी पठाईएको व्यहोरा अनुरोध छ।

नागेश्वर मंडल
वडा अध्यक्ष

दर्ता नं.:- 666
मिति:- 2078/06/28

Photo 1: Letter from ward no 4, Ward office, Hanumannagar Kankalini Municipality


ईलाका प्रहरी कार्यालय
शाखा

२ नं. प्रदेश
भारदह, सप्तरी नेपाल।

पत्र संख्याः-२०७८/२०७९
प्रा.प.स :-
चलानी नम्बर :- 929

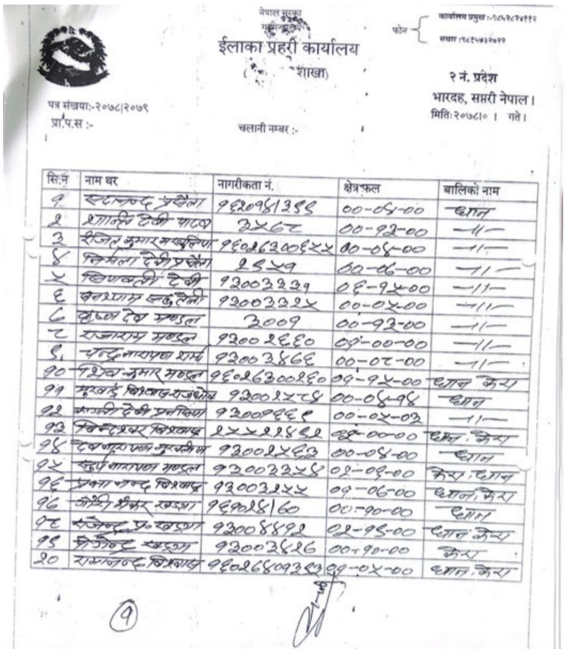
विषयः-जानकारी गराएको सम्बन्धमा।

यो जो जस संग सम्बन्ध राख्दछ।

उपरोक्त सम्बन्धमा जिल्ला सप्तरी हनुमाननगर कंकालिनि नगरपालिका वडा नं...3... .... टोल बस्ने निम्न मानिसहरुको मिति २०७८।०७।०१ गते देखि २०७८।०७।०३ गते सम्म आएको अबिरल वर्षा संगै हावाहुरी तथा नदी कटानको कारणले आएको बाढीले निम्न मानिसहरुको निम्नानुसारको खेतबारीहरुमा लगाईएको बिभिन्न अन्नबाली धान,केरा,तरकारी समेत क्षति भएको बिवरणहरु यसै पत्र साथ सम्लग्न राखी पाना 99. पठाएको व्यहोरा जानकारी गराईएको छ।

कार्यालय प्रमुख
चन्द्र प्रसाद यादव
प्रहरी नायब निरीक्षक

Photo 2: Letter from Police office, on collected data from ward no. 3, Hanumannagar Kankalini Municipality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२ नं. प्रदेश
भारदह, सप्तरी नेपाल।

सि.नं	नाम थर	नागरीकता नं.	क्षेत्रफल	बालिको नाम

Photo 3: Collected data of ward no 3 on loss and damage in the ricePhoto 3: Collected data of ward no 3 on loss and damage in the rice

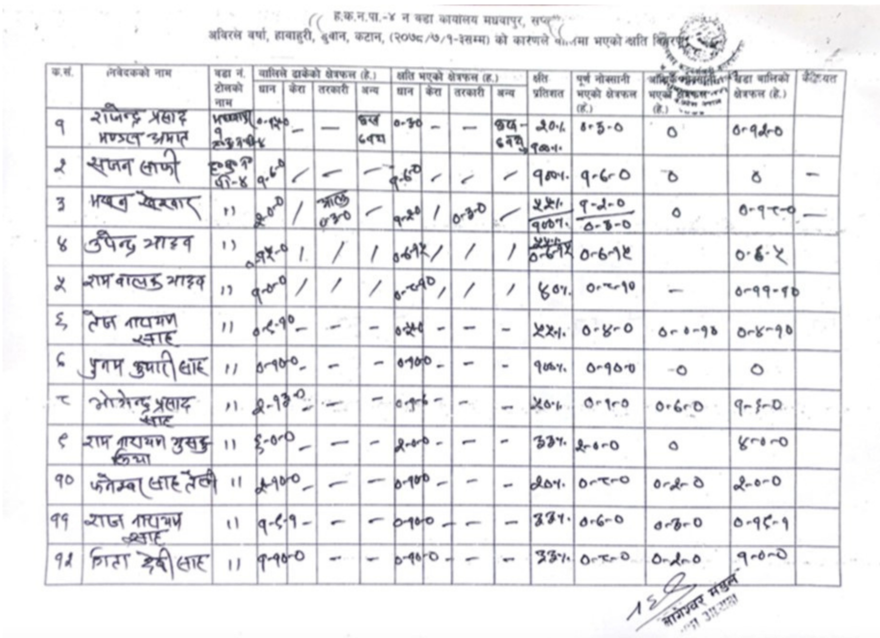
Photo 4: Collected data of ward no 4 on loss and damage in the rice and other crops

Written by: **Abhishek Shrestha, Arati Poudel and Prayash Adhikari**

DIGO BIKAS INSTITUTE

Battisputali, Kathmandu, Nepal
www.digobikas.org
+977 1 4610656 | info@digobikas.org
/digobikasinstitute | /digobikasdotorg